DENZEL MATHIBA

Full-Stack Software Developer

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GitHub: github.com/graymathiba · **Portfolio:** https://graymathiba.github.io/denzel-mathiba-portfolio
Open to remote roles · Open to immediate start · Available for contract or full-time

PROFESSIONAL SUMMARY

Full-stack developer with a National Certificate in Information Technology and a National Diploma in progress, plus 12 months of industrial attachment delivering live technical support in a production IT environment. Sole developer and founder of GraySale ERP — a local-first retail Point of Sale, inventory and double-entry accounting platform with 148 REST endpoints, a normalised 41-table SQLite schema, offline-first sync with idempotent replay, ZIMRA tax fiscalisation with ECDSA device signing and a React admin UI, built with zero runtime backend dependencies. Equally hands-on with the physical layer — PC assembly, LAN and Wi-Fi deployment, Windows/Linux administration, POS peripherals — and comfortable owning a product end to end, from schema design through CI to a packaged Windows release.

TECHNICAL SKILLS

Languages: JavaScript (ES2022), TypeScript, Python, Dart, C#, C++, SQL, HTML5/CSS3, Bash, PowerShell, Git
Frontend: React 18, React Router, Vite, HTML5/CSS3, responsive admin dashboards, Vitest, Testing Library, jsdom, browser-side ESC/POS receipt printing
Backend & APIs: Node.js 22 (ESM), Express, REST API design (/api/v1), Socket.io, WebRTC, custom HTTP router, token authentication, RBAC, multi-tenancy, rate limiting, idempotent API design
Databases: SQLite (node:sqlite, WAL), PostgreSQL (Supabase), SQL normalisation, indexing, ER modelling, 12 versioned schema migrations, backup & restore (VACUUM INTO), Firebase Auth / Crashlytics / Cloud Messaging
Mobile & Desktop: Flutter 3 / Dart 3.12, Riverpod, Hive (encrypted local-first), sqflite, dio, http, shared_preferences, workmanager background sync, flutter_webrtc, WebView2 (Windows), Win32 job objects
DevOps & Tooling: Git, GitHub Actions CI/CD, Docker, nginx, Linux (Node 22-alpine), Render, ESLint, Prettier, Husky + lint-staged, node:test, Vitest, Windows packaging & Inno Setup installers
Domain: Retail POS, ERP, double-entry accounting, inventory & stock control, purchasing, AR/AP ledgers, ZIMRA FDMS fiscalisation, licensing & entitlements, offline-first sync, multi-currency payments
IT Support & Infrastructure: Windows & Linux installation and configuration, PC assembly & hardware diagnostics, LAN/Wi-Fi deployment and troubleshooting, TCP/IP, printers / scanners / POS peripherals, on-site client support

PROFESSIONAL EXPERIENCE

Full-Stack Developer & Founder — GraySale ERP — Sep 2026 – Present

Retail POS, inventory and double-entry accounting platform · Independent product · Bulawayo, Zimbabwe · Remote / packaged for Windows

- Designed, built and shipped GraySale ERP end to end as sole developer: 148 REST endpoints across 13 route modules, a normalised 41-table SQLite schema (WAL journaling, foreign keys, 12 versioned migrations), 55 backend modules and a 63-module React 18 admin SPA — with zero runtime npm dependencies on the server.
- Implemented the full commercial domain: catalog with price tiers, FEFO batch tracking, unit-of-measure conversion and product bundles; an auditable stock-movement ledger; POS with refunds, layaway instalments, quote-to-sale, audited cash-drawer control and multi-currency cash exchange; purchasing and goods receipts; AR/AP ledgers with payment allocations; and balanced-only double-entry journal posting with trial balance, P&L, balance sheet and bank reconciliation.
- Engineered an offline-first sync for unreliable connectivity: client-generated UUIDs keyed to a unique index for idempotent replay, server-side re-pricing with price-drift reporting, per-sale failure isolation and capped batches — the Flutter till writes each sale to local SQLite before any network call and drains its queue every 15 minutes in the background via Workmanager.
- Hardened the platform for production: scrypt password hashing with per-user salts and timing-safe comparison, SHA-256-only session token storage with 12-hour TTL and single-session enforcement, login rate limiting (5 failures / 5 minutes), a role-to-capability matrix, strict tenant isolation and an audit log on sensitive actions.
- Integrated ZIMRA FDMS tax fiscalisation (device registration, open/close day, receipt submission) using ECDSA P-256 device keys and a hand-built PKCS#10 CSR, with a full offline simulator and fiscal queue so a taxed sale is never lost when connectivity drops; also verified EcoCash payment webhooks by HMAC and RSA signature.
- Built a vendor licensing platform and Control Tower: product-key issuance and redemption across monthly, quarterly, annual and lifetime plans; tenant block, extend and revoke; key-inventory and platform dashboards; and live “connected shops” presence tracking.
- Delivered the client-facing layer: a browser-side ESC/POS thermal print engine (Web Serial → WebUSB → print fallback, 80/58 mm, cash-drawer pulse), WhatsApp and email receipt sharing with no server round-trip, and A4 quote, invoice and report document generation.
- Built operational resilience and release engineering: scheduled SQLite VACUUM INTO backups with rotation and one-click restore, storage and licensing gates, hardened security headers and path-traversal containment on every response, 29 test files (node:test + Vitest), ESLint / Prettier / Husky pre-commit hooks, and a GitHub Actions CI pipeline running format-check, lint, dependency audit, tests with coverage and build.
- Packaged for Windows end to end: a release script bundles the Node runtime, server and built SPA into a distributable install, launched by a Flutter desktop host that starts the bundled server in an embedded WebView2 window with a 60-second health probe and Win32 job-object teardown.

Information Technology Intern — BeamLink Tech **Aug 2025 – Jul 2026**

Work-Related Learning (industrial attachment) · Bulawayo, Zimbabwe · On-site enterprise client support

- Delivered on-site technical support for enterprise clients — Bata Zimbabwe, Village Lodge, Africa Mall, Diamond Chicken and Midlands State University — covering network deployment, POS maintenance and hardware provisioning against strict zero-downtime targets in live retail and hospitality environments.
- Installed wired and wireless network nodes for a high-traffic hospitality site, debugged enterprise-grade POS in a busy mall, and resolved real-time IT failures in a high-volume food outlet under a “5 nines” uptime expectation.
- Contributed to the company's internal ERP — applied SQL normalisation and schema optimisation across sales, inventory and client datasets, diagnosed and fixed production bugs, and scoped custom ERP modules against client requirements.
- Assembled, configured and stress-tested PCs and high-performance workstations for Bulawayo Polytechnic and corporate customers, covering driver installation, thermal-paste and power-supply / RAM diagnostics, and performance tuning.
- Worked across the digital-to-physical production pipeline — raster image processing, colour matching and large-format inkjet and CNC cutting output — translating digital layouts into precision physical products.
- Cleaned, analysed and visualised real operational, sales and inventory datasets and automated recurring data tasks with Python and Excel, surfacing discrepancies that informed management decisions; also authored technical documentation and standard operating procedures.

Freelance Digital Marketer & Affiliate Manager — Self-employed **2022 – Present**

Remote · Paid client work · Ongoing alongside studies and employment

- Developed and executed affiliate marketing strategies that drove traffic and increased sales for clients.
- Managed partnerships with online platforms, optimised campaigns and analysed performance against ROI.
- Produced engaging promotional content and maintained strong working relationships with affiliates and stakeholders.

PROJECTS

GryChat — Secure P2P Messaging Platform — Final-year Diploma project **2026**

Full-stack, live on Render · TypeScript / Express / Socket.io backend · Flutter client for Android, iOS and Windows

- Built a full-stack messaging platform: a TypeScript + Express + Socket.io backend with a Flutter client across Android, iOS and Windows, using Firebase Authentication, Crashlytics and Cloud Messaging alongside Supabase PostgreSQL and Storage with row-level security.
- Implemented peer-to-peer transport using flutter_webrtc with a lightweight Socket.io signalling layer, and a local-first architecture using encrypted Hive storage so the app remains fully functional offline.
- Shipped 1:1 and group chat with quoted replies, emoji reactions, typing indicators, voice notes and media playback; fixed a live group-routing defect and added group-membership validation, sanitised reply persistence and reaction persistence to the database.
- Deployed with Docker (Node 22-alpine), nginx reverse proxy and Render; packaged a Windows installer with Inno Setup; CI runs flutter analyze, flutter test (18 tests) and a backend tsc build plus node:test suite (11 tests).
- Hardened the platform with Firebase token verification on every socket handshake, crash reporting and published privacy policy and terms of service.

EDUCATION

National Diploma in Information Technology — Bulawayo Polytechnic **2024 – Present**

In progress — HEXCO-accredited. Modules include Software Engineering, Database Administration, Network Administration, Hardware Administration, Object-Oriented Programming and Entrepreneurial Skills Development.

National Certificate in Information Technology — Bulawayo Polytechnic **2024**

Modules include Computer Systems Maintenance, Programming Concepts, Database Concepts, Computer Networking, Computer Security and Safety, Health, Environment & Quality Skills.

A-Levels — Hamilton High School, Bulawayo **2021 – 2022**

ZIMSEC Ordinary Level (9 subjects) — Sizane High School, Bulawayo **2016 – 2019**

Nine subjects at grade C or better, including A grades in Mathematics and Ndebele, plus Combined Science, Physics, Chemistry, Biology and Computer Science.

CERTIFICATES & PROFESSIONAL DEVELOPMENT

- HEXCO Individual Statement of Results — National Certificate (2024) and National Diploma (2025), Bulawayo Polytechnic.
- Nine O-Level subjects at grade C or better, including A grades in Mathematics and Ndebele (ZIMSEC).

LANGUAGES

English (professional working proficiency) · **Ndebele** (first language) · **Shona** (native)